

Sushma Sindhe

AI/ML & Generative AI Engineer | NLP | LLMs | MLOps | AWS | Python

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SUMMARY

AI/ML Engineer with 4+ years of experience in designing, developing, and deploying scalable machine learning and NLP solutions across finance and enterprise domains. Skilled in **Generative AI** (LLMs, Prompt Engineering, RAG, LangChain, Vector Databases like Pinecone/FAISS) and **transformer based models** (BERT, T5, GPT). Experienced in **end to end ML pipeline development** from data preprocessing and model training to deployment and monitoring using **FastAPI, Streamlit, Docker, Airflow, and CI/CD**. Proficient in **Python, NumPy, Pandas, Scikit learn, TensorFlow, and PyTorch**, with strong expertise in MLOps and cloud platforms such as **AWS, IBM Cloud, and Snowflake**. Adept at optimizing model performance, reducing latency, and improving accuracy in production-grade AI systems.

EXPERIENCE

Generative AI Engineer, PNC, United States

Jan 2024 — Present

- Designed and fine-tuned transformer based LLMs (BERT and T5) and deep neural networks for customer service automation, achieving a 27% reduction in average response time and enhancing resolution accuracy.
- Facilitated in building prototype text generation models using LSTM and GRU architectures, improving training efficiency by 20% on small to mid-scale datasets during early model experimentation.
- Developed chatbot frameworks for local business clients, enabling automated customer interactions and reducing manual support tasks by over 30%.
- Fine-tuned pre-trained transformer models (GPT-2, T5) on domain-specific corpora for document summarization and content generation tasks, enhancing output relevancy by 25%.
- Performed data cleaning and annotation for NLP datasets across three projects, increasing labeled data accuracy by 15% to enhance supervised learning model performance.
- Collaborated with senior developers to containerize ML models using Docker and Flask APIs, supporting faster testing and deployment in staging environments.

Machine Learning Engineer, CDK Global, India

Jul 2021 — Jul 2023

- Built and deployed a scalable NLP pipeline to analyze 2M+ financial documents monthly, reducing manual processing time by 45% and improving data extraction accuracy by 18%.
- Led efforts to define and prioritize machine learning applications, resulting in the deployment of three predictive models that increased fraud detection rate by 22%.
- Trained and optimized ensemble models (XGBoost, LightGBM) for credit risk prediction, reducing false positives by 16% and improving model F1-score by 12%. Deployed real-time model monitoring tools using Prometheus and Grafana, enabling anomaly detection and model drift alerts with under 5-minute latency.
- Conducted A/B testing for model updates, resulting in a 12% increase in customer engagement through personalization algorithms.
- Documented all pipelines, APIs, and model architecture in compliance with internal audit standards, improving audit turnaround time by 35%.
- Optimized cloud resource allocation for ML workloads, reducing AWS inference costs by 18% while maintaining SLA requirements.
- Experience in database schema design and relational data modeling to support ML applications.

Data Science Intern, DXC Technology, India

Feb 2021 — Jul 2021

- Integrated MLOps practices using Docker, Airflow, and CI/CD pipelines, cutting model deployment time by 40% and ensuring consistent version control and rollback capabilities.
- Integrated basic model monitoring scripts using Python logging and API usage tracking, improving early detection of runtime errors and maintaining more than 95% model uptime.
- Built text pre-processing modules (tokenization, stopword removal, lemmatization) as reusable Python functions, reducing pre processing time by 40% across projects.
- Documented key model architecture decisions and hyperparameter tuning logs, improving reproducibility and reducing onboarding time for new team members by 30%.

EDUCATION

University of New Haven Connecticut, United States - *Master's in Data Science*

Jan 2023 — Aug 2024

CMR Institute of Technology, India - *Bachelor's in Electronics and Communication Engineering* Jul 2017 — Jun 2021

SKILLS

Programming	: Python, SQL, UNIX Shell Scripting, Bash, TypeScript, R, Java, C++
Big Data & Data Engineering	: ETL Pipelines, Data Lake Architecture, Data Quality Automation, Apache Spark, Apache Kafka, Apache Hadoop, Apache Airflow, SAS, Talend

Machine Learning & Generative AI	: Deep Learning, NLP, LLMs (BERT, GPT, T5), Generative AI (Prompt Engineering, RAG, LangChain, Pinecone/FAISS), Predictive Modeling, Feature Engineering, Statistical Analysis, Model Deployment (FastAPI, Streamlit, Docker, Kubernetes, REST APIs), MLOps (CI/CD, AWS SageMaker, Airflow), Neural Networks
Cloud & Platforms	: AWS (S3, Lambda, Redshift, Glue, Lake Formation, CodePipeline), IBM Cloud, Snowflake, Jenkins, AWS Systems Manager, AWS Step Functions, AWS CodePipeline, AWS CodeDeploy, Git, Kubernetes, Agile/Scrum
Visualization & BI	: Tableau, Power BI, Amazon QuickSight, Looker